CS455

AI Engineering Log

CodeClash Deliverable 3

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`https://github.com/SrujanBhirud/CodeClash`

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These entries are the interactions in which a recommendation had to be decided for Deliverable 3. Other sessions did ordinary drafting, and those are not included. A prompt is the request as sent. A reply is summarised. Each entry then records the evaluation, the error, the change, the artifact, and the reason.

1 Sequence

Table 1: Entries in order.

No.	Stage	Date	Decision
L1	On main	5 Nov	The compile directory is writable. The run stays unprivileged.
L2	On main	5 Nov	The hardening agent may read and run. It may not publish, edit, or score.
L3	On main	5 Nov	Signup mail is a 6-digit code. The long token is only the local fallback.
L4	Not launched	5 Nov	One VM is the host shape. This account would be billed, so none was started.
L5	Local drafts	5 Nov	Three suites start from the sample. The closed-suite report is discarded.

2 Entries

2.1 L1. Sandbox permission

On main · 5 November 2026 · Om Chandrakant Chaudhari, reviewed by the team · Modified

1. Prompt

Compiled languages fail in CI. The sandbox drops every capability. Find the smallest change that lets the compiler write the binary, and keep the run unprivileged.

2. Reply (summarised)

The model first suggests putting `CAP_DAC_OVERRIDE` back on the compile container. The follow-up, after the failing test is shown, is to leave capabilities dropped and make the private temp directory mode `0777` before compile, then mode `0755` on the contents after a successful compile, so user 65534 can execute.

3. Decision

- **Evaluation.** The directory mode fixes the compile error without handing the container a capability.
- **Error.** Restoring `CAP_DAC_OVERRIDE` would undo the drop-all rule for every compile, including a hostile one.
- **Change.** `openWorkDir` chmod's the temp directory, then the compiled files. The run stays `-user 65534` and `/work` read-only.

- **Artifact.** `judge/src/runner.ts`. Jira CC-53. Pull request 47.
- **Reason.** C, C++, Java, and Go then accept a sum program. A fork bomb is still contained as RE or TLE.

2.2 L2. Agent boundary

On main · 5 November 2026 · Om Chandrakant Chaudhari, reviewed by the team · Kept

1. Prompt

Settle the test-hardening agent so a demo can show a result twice. The model is optional. Publish, edit, and score stay denied.

2. Reply (summarised)

The model proposes a second agent that may publish a repaired test when the score is low. It also proposes `gemini-2.5-flash`. A probe of the key returns 404 for that model and 200 for `gemini-3.6-flash`. With thinking left on, the call exceeds the old 20-second timeout. With thinking off and a 60-second limit, the call returns text.

3. Decision

- **Evaluation.** One worker, one tool list, and a saved run are enough for the slot.
- **Error.** A publish tool, and a model name the new keys cannot call.
- **Change.** Tools may read the problem and run the sandbox. A `publish` call is denied and audited. The model is `gemini-3.6-flash` with thinking off. Keys stay in gitignored env. `scripts/demo-harden.mjs` writes `Deliverables/demo/hardening.json`. A second run with `AGENT_PROVIDER` unset writes `hardening-empty.json` and finishes in seconds. The slot opens the saved file.
- **Artifact.** `agent/src/engine/ai.test.ts`, `scripts/demo-harden.mjs`, the two JSON files.
- **Reason.** The deny is a test, not a slide. The empty-provider file is what we repeat if the model is slow.

2.3 L3. Six-digit mail

On main · 5 November 2026 · Om Chandrakant Chaudhari, reviewed by the team · Modified

1. Prompt

Signup in the demo should be the email code we already fought with, not a long token pasted from the page.

2. Reply (summarised)

The model keeps the long token and adds a mailer that also prints it. The follow-up is the code shape already used in the other tree: six digits, fifteen minutes, Mailpit on SMTP 1025.

3. Decision

- **Evaluation.** A code a person can type is the thing the slot can show. Mailpit is local and costs nothing.
- **Error.** Returning the long token whenever mail is configured trains the demo on a path production hides.
- **Change.** `newVerifyToken` is a six-digit code, valid for fifteen minutes. The token is returned only when SMTP is unset and `NODE_ENV` is not production. Auth tests still post whatever the API returns.

- **Artifact.** `server/src/domain/product.ts`, `server/src/mail.ts`, `client/src/pages/Auth.tsx`, Mailpit in `docker-compose.yml`.
- **Reason.** A local register returned `mailed: true`, Mailpit held the code, and `POST /api/auth/verify` returned 200.

2.4 L4. One VM

Not launched · 5 November 2026 · Om Chandrakant Chaudhari, reviewed by the team · Rejected for this account

1. Prompt

The live demo runs on one AWS VM. This PC is the backup. Do not add a NAT gateway, a load balancer, or RDS. If a `t3.micro` would bill, stop.

2. Reply (summarised)

The model lists a public `t3.micro`, 20 GB gp3, ports 22 and 80, Caddy, and three systemd units. It treats a missing free-tier row as "still free" and offers to launch.

3. Decision

- **Evaluation.** The shape is right: one machine, Mongo and Redis on localhost, the judge on that machine's Docker socket.
- **Error.** `get-free-tier-usage` on 5 November lists only AWS Glue. EC2 is not in the 12-month allowance. Launching would bill.
- **Change.** No instance was started. `deploy/README.md` keeps the commands, including `aws ec2 stop-instances`, for an account that still has the allowance. The ten-minute script uses `http://127.0.0.1:5173`.
- **Artifact.** `deploy/Caddyfile`, the three unit files, `Deliverables/demo-script.md`.
- **Reason.** Actual cloud spend stays $0. A billed instance left running would exhaust the course cap for no demo that this PC cannot already run.

2.5 L5. Building a suite from the sample

Local drafts · 5 November 2026 · Om Chandrakant Chaudhari, reviewed by the team · Modified

1. Prompt

Document the hardening pipeline on harder, average problems already in the database. Include the mutant source, what was discarded, and which test was hardened. Then: scratch that. Take three problems, start their tests, and rerun the pipeline as the tests grow, so the demonstration shows the suite being built.

2. Reply (summarised)

The first reply runs five medium problems whose suites are already complete: Climbing stairs, Balanced brackets, Maximum subarray, Pair with a given sum, and Range sum. Every wrong program is already killed. The score does not move, and no test is added. The write-up says the suites were finished.

3. Decision

- **Evaluation.** A closed suite can show that a wrong program is held. It cannot show a test being added.
- **Error.** Reporting "nothing to add" as the result of the bot.

- **Change.** Three new drafts, copied from Larger of two, Balanced brackets, and Maximum subarray, each reduced to the sample. The next tests are the short lines the run asks for, plus the obvious neighbour. The random triple and the 64,000-character strings are not saved. The published problems are not edited.
- **Artifact.** `Deliverables/demo/hardening-build.md` and `hardening-build.json`. The pipeline figure and both provider paths are in Deliverable 3, §the test-hardening bot.
- **Reason.** On Larger of two the run names `-100 -50`. On brackets it names `(`, then `())((`. On the subarray, two short cases replace the random triple and the score goes from 0 to 1. The second run is the evidence that the line from the first run already catches the bug.